

EAIF: AI for Finance, Supervised Machine Learning 3

Flipped-Class Activity: Big Data, New Tricks for Econometrics

Work in small groups to replicate one piece of Hal Varian's paper 'Big Data: New Tricks for Econometrics' (Journal of Economic Perspectives, 2014) in a Colab notebook: the growth-regression example of Section 6.3, in which four variable-selection methods are compared on a dataset of 72 countries.

Before class (about 60 minutes)

- Read Varian's paper (JEP 2014, link: <https://www.aeaweb.org/articles?id=10.1257/jep.28.2.3>). Read Sections 3, 6 (especially 6.3), 8.1 and 9 carefully; skim the rest.
- Write a short answer (three to five sentences each) to these three questions and bring it to class:
 1. In the growth-regression table of Section 6.3, which variables do all four methods agree on, and why does Varian say the dataset is too small to settle what matters for growth?
 2. What is the difference between the prediction goal of Section 3 and the causal question of Section 8.1? Explain it with Varian's police-and-crime example.
 3. What does Varian mean by model uncertainty in Section 9, and why does he think averaging over models tends to predict better than choosing one?
- Open the starter notebook once before class to check that it runs:
https://colab.research.google.com/github/umatter/EDFB/blob/main/activity/Varian_Replication_Starter.ipynb

In class

1. Check-in (10 min)

- Compare your answers to the three pre-reading questions in your group. Agree on one sentence per question that you would defend.

2. Replicate (60 min)

- Open the starter notebook in Google Colab (link above) and save a copy in your Drive.
- The data (FLS-data.csv, the Fernandez, Ley and Steel 2001 growth data used in Section 6.3) load with one line from the course repository:
<https://raw.githubusercontent.com/umatter/EDFB/main/activity/FLS-data.csv>. The original replication package is on OpenICPSR (project 113925) for reference; you do not need to download it.
- Step by step:
 1. Load and inspect the data and read the codebook cell.
 2. Fit the lasso: which variables survive, and in which order do they enter the path? Compare with the BMA ranking in the column order.
 3. Fit OLS with all 41 regressors and rank effects by t-statistic. Watch what happens to a model with 42 parameters and 54 rows.

4. Fit a random forest as the extension Varian did not run. Compare the three rankings and the three test errors next to the naive baseline.

- You can use AI tools for code help, but make sure you understand what is happening in every cell you keep.

3. Wrap-up (20 min)

- Each group answers the three reflection questions at the end of the notebook in text cells, now with its own numbers:

- Where the methods agree and where they diverge, and whether the table supports Varian's 'too small to resolve'

- Whether the exercise answered a prediction or a causal question

- Where model uncertainty showed up in the results

- Two groups present their answers; the class compares numbers across groups (different splits give different rankings, which is the point).

Deliverable

The completed starter notebook with the three reflection answers in text cells, submitted via Moodle by the end of class. The deliverable is formative: it is not graded, but it is required, and the feedback feeds into the group assignment.